“The Great Dying”

The most dramatic mass extinction took place at the end of the Permian period, 252 million years ago. Also known as “The Great Dying,” it resulted in the loss of up to 94 percent of marine species and 70 percent of terrestrial vertebrate families. For example, it led to the extinction of the last of the trilobites.

Potential causes for the mass extinction include intense volcanic activity, which led to global warming, ocean acidification, and acid rain. This was one of the biggest periods of volcanic activity in Earth’s history. Lasting nearly 1 million years, the eruptions flooded more than 0.8 million sq miles (2 million sq km) of ancient Siberia with basalt lava.

“Modeled future extinction rates are projected to be 10,000 times Earth’s historical geological background rate.”

Ron Wagler

American academic

Phased losses

All life today is descended from the small minority of species that remained at the start of the Triassic period. Another mass extinction event occurred across two distinct phases at the end of the Triassic period (200 million years ago), wiping out around 80 percent of species. A potential cause may have been the extensive volcanic activity across the Central Atlantic Magmatic Province, which led to climatic changes. In the seas, many reptiles, cephalopods, mollusks, and reef-building organisms died out. On land, most of the reptile-like archosaurs and many large amphibians became extinct. The loss of the archosaurs, in particular, opened up ecological niches that the dinosaurs would fill.